

CYPRUS INTERNATIONAL UNIVERSITY



Electrical - Electronic Engineering - English - Undergraduate SIGNALS & SYSTEMS 2025-26 Spring Course Syllabus

Code(s)	Course Name	Type	Weekly Course Hours			Credits	ECTS
			T	P	L		
CMPE321 EELE321 ELE307 EEE321	SIGNALS & SYSTEMS	Course	4	0	1	4	7
Prerequisite		MATH201		Language of Instruction		English	
Course Lecturers / Assistants		Asst. Prof. Dr. Mustafa MULLA, Res. Asst. Umar Bindir AHMED					
Textbooks and/or References							
Book / Reference 1	Oppenheim, A. V., Willsky A. S. with Nawab, S. H., Signals & Systems, 2nd Edition, Prentice-Hall, 1997.						
Book / Reference 2	Simon Haykin & Barry Van Veen, Signals and Systems, 2nd edition, John Wiley & sons, Inc 2003.						
Book / Reference 3	Buck, J. R., Daniel, M. M. and Singer, A. C., Computer Explorations in Signals and Systems Using MATLAB, Prentice-Hall, 1997.						
Book / Reference 4	Kwakernaak, H., Sivan, R., Modern Signals and Systems, Prentice-Hall, 1991.						
Course Description							
Classification of Signals and Basic Signal Properties. Time Domain Models of Linear Time Invariant (LTI) Systems: Continuous time systems. Causal LTI systems described by differential equations. System block diagrams. The solutions of differential equations. The unit impulse response and convolution integral. State variable analysis of LTI systems. Discrete time systems. The unit sample response and discrete convolution. Fourier series and Fourier transform representation of continuous-time and discrete- time periodic signals. Time and frequency characterization of signals and systems. Z-transform and inverse z-transform. Region of convergence of the z-transform. z-domain analysis of discrete LTI systems. LTI Systems With Random Inputs. Definition of Random variables, stochastic process, first and second order statistics, moment, correlation and co-variance, stationary process, ergodicity. System resonse.							
Weekly Course Plan							
Week 1-8	Topics			Week 9-15	Topics		
Week 1	Course introduction. Basic CT and DT signals, time-frequency domains introduction. Transformation of the independent variable, signal energy and power.			Week 9	Difference and differential equation representations of LTI systems. Block diagram representations.		
Week 2	Course introduction. Basic CT and DT signals, time-frequency domains introduction. Transformation of the independent variable, signal energy and power.			Week 10	Fourier representations for four classes of signals.		
Week 3	Classification of Signals. Elementary Signals. Basic operations on signals.			Week 11	Fourier representations for four classes of signals.		
Week 4	CT and DT systems. Basic system properties. Linear Time-invariant systems			Week 12	Fourier representations for four classes of signals.		
Week 5	DT-LTI systems Convolution sum.			Week 13	Properties of Fourier Representations.		
Week 6	CT-LTI systems convolition integral.			Week 14	Properties of Fourier Representations.		
Week 7	MIDTERM EXAMS			Week 15			
Week 8	Properties of LTI systems. Interconnection of LTI systems. Relations between LTI system properties and the impulse response.			-	-		
Evaluation Tools				Weight in Total (100%)			
Midterm - Midterm				35			
Final - Final				45			
Quiz - Quiz				10			

Assignment - Lab	10
It is obligatory that students attend to their classes. Absenteeism of students shall be monitored by the relevant teaching staff. It is imperative that students attend minimum of 70% of class hours in their courses.	

#	Learning Outcomes	SOLO Taxonomy
LO1	Identify the properties of signals,	2
LO2	Apply the Discrete Time and Continuous Time operations of signals and illustrate the results,	3
LO3	Characterize systems depending on their properties.	3
LO4	Diagram and analyze a systems block diagram.	3
LO5	Derive the output signal of a system for a given input.	4
LO6	Derive the transformation of signals from time domain to frequency domain or vica versa,	4
LO7	Implement signals and systems by using a numerical modelling software (MATLAB).	4
Solo Taxonomy Average		3.29

#	Program Outcomes
PO1	Should have sufficient knowledge in mathematics, science, and subjects specific to the relevant engineering discipline.
PO2	Should have the ability to use theoretical and applied knowledge in mathematics, science, and related engineering disciplines in complex engineering problems.
PO3	Should have the ability to detect, define, formulate, and solve complex engineering problems.
PO4	Should have the ability to select and apply appropriate analysis and modeling methods to solve complex engineering problems.
PO5	Should have the ability to design a complex system, process, device, or product to meet specific requirements under realistic constraints and conditions.
PO6	Should have the ability to apply modern design methods.
PO7	Should have the ability to develop, select, and use modern techniques and tools necessary for the analysis and solution of complex problems encountered in engineering applications.
PO8	Should have the ability to use information technologies effectively.
PO9	Should have the ability to design experiments, for the study of complex problems or discipline-specific research topics.
PO10	Should have the ability to conduct experiments, collect data, analyze and interpret results for the study of complex problems or discipline-specific research topics.
PO11	Should have the ability to work in intradisciplinary teams.
PO12	Should have the ability to work in interdisciplinary teams.
PO13	Should have the skills to work individually.
PO14	Should have the ability to communicate effectively verbally and in writing.
PO15	Should have the knowledge of at least one foreign language.
PO16	Should be able to write effective reports, understand written reports, and prepare design and production reports.
PO17	Should have the ability to make effective presentations.
PO18	Should have the ability to give and have clear and understandable instructions.
PO19	Should gain consciousness (awareness) about the necessity of lifelong learning.
PO20	Should have the ability to access information.
PO21	Should have the ability to follow developments in science and technology and constantly renew himself/herself.
PO22	Should gain the awareness of professional and ethical responsibility and should act in accordance with ethical principles.
PO23	Should gain knowledge about the standards used in engineering applications.
PO24	Should gain knowledge about project management, risk management, and change management practices in business life.
PO25	Should gain awareness about entrepreneurship, and innovation.
PO26	Should gain knowledge about development in sustainability.
PO27	Should gain knowledge about the effects of engineering practices on health, environment, and security at universal and social dimensions and the problems of the age reflected in the field of engineering.
PO28	Awareness should be gained about the legal consequences of engineering solutions.